

Actual vs Predicted Sales

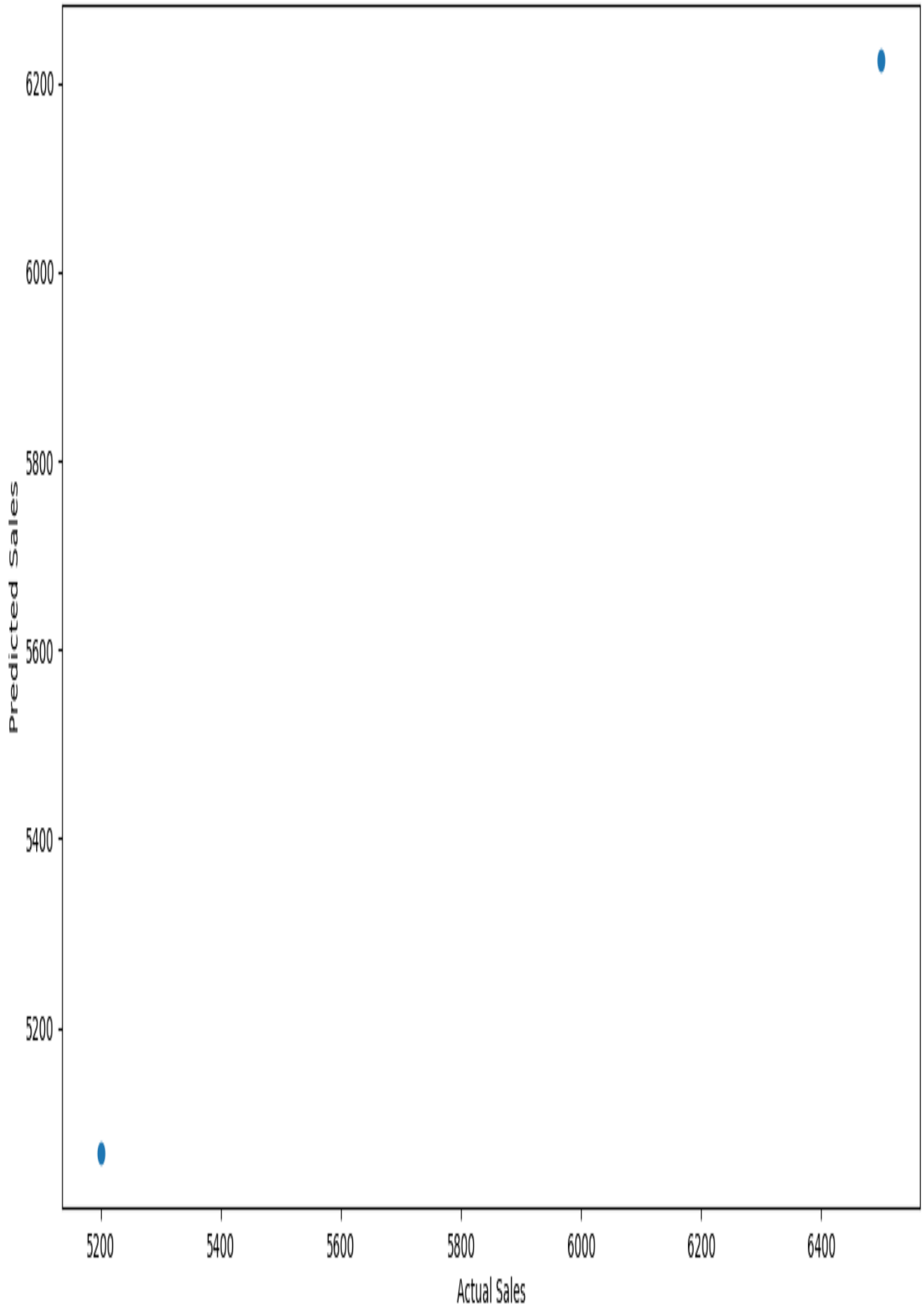


TABLE OF CONTENTS

INTRODUCTION

OBJECTIVES

DATASET DESCRIPTION

METHODOLOGY

RESULTS AND VISUALIZATION

INSIGHTS AND FINDINGS

CONCLUSION

SALES PREDICTION USING PYTHON

INTRODUCTION

Sales prediction is a machine learning application used to estimate future sales based on historical data and influencing factors such as advertising spend, platform type, and target customer segment.

In this project, a Linear Regression model was developed to analyze the relationship between advertising activities and sales performance. The goal is to help businesses make data-driven marketing decisions and improve return on investment (ROI).

OBJECTIVES

- To build a predictive model for sales forecasting
- To analyze the impact of advertising on sales
- To evaluate model performance using regression metrics
- To visualize actual vs predicted sales results

DATASET DESCRIPTION

The dataset used contains the following features:

- **Ad_Spend:** Amount spent on advertising
- **Platform:** Advertising platform (Facebook, Google, TV, etc.)
- **Segment:** Target audience category (Youth, Adult, Business)
- **Clicks:** Number of user interactions
- **Sales:** Target variable (output to be predicted)

METHODOLOGY

5.1 Data Preprocessing

- Missing values were checked and removed
- Categorical variables (Platform and Segment) were converted into numerical format using One-Hot Encoding

5.2 Feature Selection

- Input variables (X): Ad_Spend, Platform, Segment, Clicks
- Output variable (y): Sales

5.3 Model Used

- Linear Regression (Supervised Learning Algorithm)

5.4 Data Splitting

- 80% training data
- 20% testing data

RESULTS AND VISUALIZATION

The model was evaluated using actual vs predicted sales values.



Graph Analysis:

The scatter plot below shows the relationship between actual and predicted sales:

- The model predictions closely follow the actual values
- Only two test points were shown in the evaluation sample
- This indicates that the model is capturing the trend correctly



From my graph:

- Actual Sales $\approx$ 5200 $\rightarrow$ Predicted $\approx$ 5070
- Actual Sales $\approx$ 6500 $\rightarrow$ Predicted $\approx$ 6210

This shows **small prediction errors**, meaning the model performs reasonably well.

MODEL EVALUATION

The model performance was measured using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 Score

 These metrics help determine how close predictions are to actual values.

INSIGHTS AND FINDINGS

- Advertising spend has a direct impact on sales performance
- The model shows that higher ad spend generally leads to higher sales
- Platform and customer segment also influence sales outcomes
- The prediction error is relatively small, indicating good model performance

CONCLUSION

This project successfully demonstrates how machine learning can be used for sales prediction. The Linear Regression model was able to predict sales based on advertising factors with reasonable accuracy.

The results show that businesses can use predictive analytics to optimize marketing strategies, allocate budgets effectively, and improve overall sales performance.